

1. Fet Timetable ~ open source software

- ❖ Institution name (litakalotokea juu ya timetable (mtu aweze kuingiza isiwe fixed)
- ❖ Days of the week
- ❖ Hours (masaa) either 24/12 per day
- ❖ All subjects (masomo yote kwa kutumia course code).
- ❖ Activity tag (if it is Seminar, lecture, practical, lab). Aina ya shughuli
- ❖ Teachers
- ❖ Students (Year/Faculty efg FASS, Group/Programs eg BsC 1, Sub groups eg BAED 1 eg EngliSh + kiswahili, BBA
- ❖ Buildings (Rooms and Capacity) eg AH 150, etc
- ❖ Activities (teacher, activity tag eg lecture, Subject, Sessions per day/week/time
- ❖ Specifying room zitakazotumika kwa ajili ya Lecture, practical, seminar, lab.
- ❖ Generate time table (Single click).

Restrictions/conditions/Scenario

✓ Time constraints

○ Teacher

- Individual teacher
- All teacher

- Max days per week
- Min day per week for a teacher or all teachers
- Max gap per day for a teacher
- Max gap per week for a teacher
- Max hours daily for a teacher
- Max span per day for a teacher
- Max hours daily with an activity tag for a teacher
- Min hours daily for a teacher
- Min hours daily with an activity tag for a teacher
- Max hours continuous for a teacher
- Max hours continuous with an activity tag for a teacher
- Min gap btn an ordered pair of activity tag for a teacher
- A teacher works in an hourly interval max days per a week
- Min resting hours for a teacher

Note: *These conditions can be applicable to all.*

○ Student

- Max days per week for a student set

- Max gaps per day for a student set
- Max gaps per week for a student set
- A student set begins early
- Max hours daily for a student set
- Max span per day for a student set
- Max hours daily with an activity tag for a student set
- Min hours daily for a student set
- Min hours daily with an activity tag for a student set
- Max hours continuously for a student set
- Max hours continuously with an activity tag for a student set
- Min gap b/n an ordered pair for a activity tags for a students set
- A student set works in an hourly interval max daily per week
- Min resting hours for a student set

Note: These conditions can be applied to all students

○ Activity

- An activity has a preferred starting time
- An activity has a set of preferred time
- An activity has a set of preferred time slot
- An activity has a set of preferred starting times
- An activity has a set of preferred time slot
- An Sub activity has a preferred starting times
- An Sub activity has a preferred time slots
- Min days b/n set of activities
- Max days b/n set of activities
- An activity ends student day
- A set of activity end student day
- A set of activity has the same starting time
- A set of activity has the same starting day
- A set of activity has the same starting hours
- A set of activity occupied
- T
- Two activities are ordered
- Two activities are ordered ...

- Two activities are consecutive
- Two activities are grouped
- Three activities are grouped
- A set of activities are not overlapping
- A set of activity tag are not overlapping
- Max simultaneous activities from a set in selected time slot
- Min simultaneous activities in a set in selected
- Min

✓ Space constraints

- Room
 - ❖ A room not available time
- Teacher
 - ❖ A teacher has a home room
 - ❖ A teacher a set of home room
 - ❖ Max change a day per teacher
 - ❖ Max room changes per week for a teacher
 - ❖ Min gaps btn room changes for a teacher
 - ❖ Max building changes per day for a teacher
 - ❖ Max building changes for a week for a teacher
 - ❖ Min building btn
 - ❖ Max room changes per day for all teacher
 - ❖ Max room changes per day for all teacher
- Student
 - ❖ A set of student has home room
 - ❖ A set of student has a set of home room
 - ❖ Max room changes per a day for a students
 - ❖ Max room changes per a Week for a students
 - ❖ Min gap btn Room
 - ❖ Max room changes for a student set
 - ❖ Max building changes per a day for a student set
 - ❖ Max building changes per week for a student set
 - ❖ Min gap btn building changes for a student set

2 -last (all students)
- Subject
 - ❖ A subject has preferred room
 - ❖ A subject has a set of preferred rooms

- Activity tag
 - ❖ An activity tag has a preferred room
 - ❖ Activity tag has a set of preferred rooms
- Combination btn subject and activity tag
 - ❖ A space + activity tag has a preferred room
 - ❖a set of preferred rooms
- Activity
 - ❖ A set
 - ❖ .. a set of preferred rooms
 - ❖ Consecutive
 - ❖ Occupies max different room

Haiwezi kuedit somo mojamoya

Tengeneza template ya kuweza kuingiza

Kutengeneza ya EU/Sup/

E-Timetable algorithm